

World IT Spending 2002-2005

Timing the Recovery

April 2002



World IT Spending 2002-2005

Timing the Recovery

April 2002

Aberdeen Group, Inc.

Aberdeen Group is a leading IT market analysis and positioning services firm that helps Information Technology vendors establish leadership in emerging markets.

Aberdeen Group performs specific projects for a select group of domestic and international clients. Each project requires a combination of strategic advice and pragmatic, experience-based action plans. Assignments range from corporate and product positioning and organizational planning to in-depth market segment research. Aberdeen consults on mergers and acquisitions, corporate positioning and investor relations, and transaction-processing benchmarks; in addition, it offers special expertise in software and midrange computer markets.

In carrying out assignments, Aberdeen uses a proprietary, comprehensive, analytical framework providing fresh insight into the complex future of computing and communications, both from technological and market-dynamics perspectives. Aberdeen also offers retainer-fee programs to a group of continuing clients.

Aberdeen principals and staff are recognized industry figures with hundreds of years of combined high-tech industry, research, and financial community experience among them. They are quoted extensively in industry trade and business publications. Each staff member is a frequent conference and seminar speaker.

In addition to client-related research and consulting, Aberdeen publishes several periodicals, Aberdeen Viewpoints, and Profiles, which summarize its analysis and research findings.

To provide us with your feedback on this research, please go to www.aberdeen.com/feedback.

The information contained in this publication has been obtained from sources Aberdeen believes to be reliable, but is not guaranteed by Aberdeen. Aberdeen publications reflect the analyst's judgment at the time and are subject to change without notice.

The trademarks and registered trademarks of the corporations mentioned in this publication are the property of their respective holders.

Warning: This publication is protected by United States copyright law and international treaties. Unless otherwise noted, the entire contents of this publication are copyrighted by Aberdeen Group, Inc. and may not be reproduced, stored in a retrieval system, or transmitted in any form or by any means without prior written consent. Unauthorized reproduction or distribution of this publication, or any portion of it, may result in severe civil and criminal penalties, and will be prosecuted to the maximum extent necessary to protect the rights of the publisher.

© 2002 Aberdeen Group, Inc., Boston, Massachusetts

Table of Contents

Preface.....	1
<i>Chapter One: Executive Summary</i>	2
<i>Chapter Two: Key Findings</i>	4
<i>Chapter Three: Methodology</i>	6
Assumptions.....	6
Definitions.....	7
IT Spending	7
Regions.....	7
Hardware	7
Software.....	7
Services.....	7
Currency	7
<i>Chapter Four: Global IT Spending Outlook</i>	8
Growth Estimates	10
Quarterly Growth	12
Top Country Spending.....	13
Hardware, Software, and Services	15
<i>Chapter Five: North America</i>	16
Overview	16
Spending Mix.....	17
<i>Chapter Six: Latin America</i>	19
Overview	19
Spending Mix.....	20
<i>Chapter Seven: Europe</i>	22
Overview	22
Spending Mix.....	23
<i>Chapter Eight: Middle East and Africa</i>	25
Overview	25
Spending Mix.....	26
<i>Chapter Nine: Asia and Pacific Rim</i>	28

Table of Contents

Overview	28
Spending Mix.....	30
Author Profiles	31
Hugh Bishop.....	31
Esmé Fantozzi.....	32
<i>Appendix A: Lexicon of Acronyms and Abbreviations</i>	33
<i>Appendix B: Related Aberdeen Research</i>	34

Figures

Figure 1: 2002 World Information Technology Spending	8
Figure 2: 1999-2005 World IT Spending	9
Figure 3: World and U.S. IT Spending Growth Rate	10
Figure 4: IT Spending - Quarterly Growth Rates	12
Figure 5: IT Spending in the Three Largest Markets 1999 - 2005	13
Figure 6: IT Spending — Countries Ranking 2 nd to 10 th	14
Figure 7: World Hardware, Software, and Services Spending 1999 - 2005	15
Figure 8: 1999 - 2005 IT Spending – North America	16
Figure 9: 1999 - 2005 IT Spending by Country – North America	17
Figure 10: 1999 - 2005 Hardware, Software, and Services Spending – North America	18
Figure 11: 1999 - 2005 IT Spending – Latin America	19
Figure 12: 1999 - 2005 IT Spending by Country – Latin America	20
Figure 13: 1999 - 2005 Hardware, Software, and Services Spending – Latin America	21
Figure 14: 1999 - 2005 IT Spending – Europe	22
Figure 15: 1999 - 2005 IT Spending by Country — Europe	23
Figure 16: 1999 - 2005 Hardware, Software, and Services Spending — Europe	24
Figure 17: 1999 - 2005 IT Spending — Middle East and Africa	25
Figure 18: 1999 - 2005 IT Spending by Country — Middle East and Africa	26
Figure 19: 1999 - 2005 Hardware, Software, and Services Spending — Middle East and Africa	27
Figure 20: 1999 - 2005 IT Spending — Asia and Pacific Rim	28
Figure 21: 1999 - 2005 IT Spending by Country — Asia and Pacific Rim	29

Figures

Figure 22: 1999 - 2005 Hardware, Software, and
Services Spending — Asia and Pacific Rim..... 30

Tables

Table 1: IT Spending Growth Rate - Countries Ranking 1st to 10th 11

Table 2: Quarterly IT Spending Table..... 13

Preface

In 2001, the Information Technology (IT) industry witnessed its years of double-digit growth come to a halt. Total IT expenditures (combined spending on hardware, software, networking equipment, and IT professional services) grew at a slightly negative rate in the U.S. and were flat worldwide. Aberdeen's World IT Spending 2002-2005: Timing the Recovery Research Report gives the reader comprehensive information about how the world will spend on IT through 2005. In this report, Aberdeen discloses when, and to what degree, IT spending will resume growth.

Using a unique approach grounded in macroeconomic fundamentals, Aberdeen has developed its World Technology Spending Model (WTSM) that is used to predict expenditures by region and by country and to further break down that information into any number of relevant categories. For this report, which is part of a continuing series of IT spending analysis reports to be published by Aberdeen, IT expenditures are analyzed in four categories: overall spending, hardware spending, software spending, and services spending. Aberdeen also leverages the WTSM approach to provide insights into specific vertical industries, revealing the global and regional markets for certain technologies.

This report is intended for anyone who has a vested interest in the future of IT spending. Specifically, the report targets: top-level executives of IT suppliers who want to analyze the prospects for their businesses globally; leaders of mid-level companies considering an expansion of their businesses outside of their home market; and members of the private equity community who need to know where opportunity is the greatest and where it is expected to grow. This report will also prove to be invaluable for the broader investment community in judging viability of markets to sustain IT consumption.

Chapter One:

Executive Summary

This Aberdeen Research Report presents an overview of worldwide commercial computing spending by region. It offers readers the following:

- A projected breakdown of worldwide IT spending by country and region for 2002-2005;
- Worldwide and regional technology growth projections for 2002 -2005;
- A dissection of regional spending by type (hardware versus software and services) for 2002-2005;
- A breakdown of projected spending for the top 16 countries ranked by IT expenditures;
- IT spending growth comparisons by country and region; and
- Aberdeen's key findings on the causes and implications of the data contained in this report.

This report represents the latest installment in Aberdeen's commitment to analyzing and articulating major market opportunities for IT suppliers. Over the 14 years that Aberdeen has been analyzing business uses of IT, we have noted the growing, increasingly widespread need for credible market numbers related to international market opportunities. Several factors have combined to generate recent interest. These factors include the following:

- International adoption rates of technology continue to expand as prices decline. Price declines in computers have accelerated. Declining prices and years of sustained economic growth have served to drive strong international investments in computers, software, and communications equipment.
- IT has become more than a tool to affect certain ends: It is now the medium through which much of the international economy flows. As IT becomes more central to the functioning of countries and companies, unique international, industry-specific business processes become inexorably intertwined with the operation of technology on a worldwide basis.
- The sustained level of productivity growth experienced in the U.S. (10 years) is widely acknowledged to be a function of

strong domestic investment in technology and automation, which has helped to drive a combination of falling inflation and unusually strong, peacetime growth in the United States. Using the U.S. as an investment roadmap, many regions such as Europe, Asia Pacific, the Middle East, and Latin America are pushing an increasing amount of capital expenditures toward technology investment in an effort to duplicate productivity gains seen domestically.

Chapter Two:

Key Findings

IT spending will begin to recover in 2002. In 2002, Aberdeen forecasts that world IT spending will grow at 3.1% and U.S. spending will increase by 4.3%.

Aberdeen's key findings are as follows:

- *IT spending will begin to recover in 2002.* Worldwide IT spending in 2001 grew at a meager 0.2%, while declining 0.4% in the United States. In 2002, Aberdeen forecasts that world IT spending will grow at 3.1% and U.S. spending will increase by 4.3%.
- *The real IT spending recovery will begin in the second quarter of 2002.* Aberdeen expects U.S. spending to improve 3.9% when measured against second quarter 2001 and world IT spending to increase by 2.7%.
- *The long-term growth rate will be less than that experienced in the late 1990's.* Aberdeen does not forecast a return to double-digit growth rates. Rather, we believe that for 2003 to 2005, world IT spending will increase at an annual rate of 4% to 5%, and U.S. spending will grow at an annual rate of 5% to 6%.
- *The factors contributing to excessive growth in the late 1990's are not repeatable.* These elements included corporate retooling around Y2K issues, the Internet/e-Commerce euphoria phenomenon, excessive venture capital investment in dot.coms, traditional enterprises accelerating e-business deployments to match the dot.com threat, and unsustainable IT spending in the telecom sector.
- *Long-term IT spending is gated by Gross Domestic Product (GDP) growth and corporate revenue growth.* IT spending now accounts for 3.85% of the world GDP and 4.65% of the U.S. GDP. Additionally, the U.S. Department of Commerce reports that over 65% of the U.S. GDP consists of consumer spending, making IT an even greater percentage of business spending and investment. IT spending decisions now factor heavily into total business expenses and must be budgeted accordingly.
- *As a result, sub-segments exhibiting hyper-growth will do so at the expense of others.* IT spending is no longer a tide that will raise all boats. Supplier competition over limited budget dollars will become increasingly fierce.
- *China's importance will dramatically increase.* Aberdeen expects that China will vault from the seventh largest market for

IT products and services (in 2001) to the fourth largest market by 2005, surpassing the U.K., France, and Italy.

- *Japan continues to languish.* A decade of low growth and capital investments has taken its toll. Japan lags other developed nations in current IT spending and will not return to positive growth levels until 2003.
- *The top 10 countries account for nearly 75% of world spending.* The U.S. alone accounts for 37.5% of world IT spending. Obviously, IT spending power is concentrated in a handful of markets.
- *The U.S. will lead the recovery.* Aberdeen expects that U.S. IT spending growth will exceed the world average for 2002 – 2005.
- *Software and services spending gain at the expense of hardware.* Worldwide hardware expenditures will increase a total of only 8.7% from 2001 to 2005, while software and services will increase 36% and 21%, respectively, over the same time period.

Chapter Three:

Methodology

In an effort to build a comprehensive framework to assess a country's likely spending on Information Technology, Aberdeen has used a wide range of economic and qualitative data. It should be noted that Aberdeen asserts that a country's ability to spend on IT is gated by a multiplicity of economic factors including, but not limited to: GDP, stage of economic development, industry composition, real economic growth rate, and the general state of its economic relationship with the United States.

The results of the study come from the analysis of worldwide quantitative and qualitative data, with specific breakouts for the 60 largest economies. The model used to obtain the results contained in this report is based on quantifiable macroeconomic variables. A central thesis of this work is that Information Technology spending is limited by the ability of the country's government and businesses to spend. Consistent with this approach, Aberdeen has grounded its technology spending predictions in the best available economic data. Due to the uncertain nature of worldwide economic conditions year-to-year, Aberdeen anticipates updating the information and assumptions as new macroeconomic data becomes available.

Assumptions

The following are assumptions used during Aberdeen's analysis of worldwide IT spending:

- There exists a relationship between the Gross Domestic Product (GDP) and the ability of the country's businesses and government to spend on Information Technology.
- There exists a relationship between a country's real economic growth rate and its ability to increase spending on Information Technology.
- There exists a relationship between a country's level of capital investment and its ability to spend on Information Technology.
- There exists a relationship between a country's industry segmentation and its ability to spend on Information Technology.
- There exists a relationship between a country's stage of economic development and its ability to spend on Information Technology.

- There exists a relationship between the per capita income of a company and its ability to spend on Information Technology.
- There exists a relationship between a country's previous investments in Information Technology and its future investments.
- There exists a relationship between the saturation level of IT investments in a country and its propensity for future spending.

Definitions

IT Spending

For the purposes of this report, Aberdeen defines IT spending as money spent on hardware, software, or services that are of a commercial grade. Aberdeen does not include communications equipment such as phones and video and audio equipment, although one often finds it included in other definitions. Networking equipment deployed as part of a corporate or governmental organization's IT infrastructure is included. However, networking equipment purchased by telecommunications carriers to build out their service networks is not included.

Regions

For this report, Aberdeen grouped countries by geographic region in order to aid in analysis and comparison between countries and between regions. The regions used are: North America, Latin America, Europe, Middle East & Africa, and Asia Pacific Rim (APR).

Hardware

Hardware is defined as a computer and the associated physical equipment directly involved in the performance of data processing or communications functions.

Software

Software is defined as the programs, routines, and symbolic languages that control the functioning of the hardware and direct its operation.

Services

Aberdeen defines services as a portion of IT spending to mean any consulting, support, or systems integration services associated with the implementation or optimization of hardware and software.

Currency

All figures, unless stated otherwise, are in U.S. dollars.

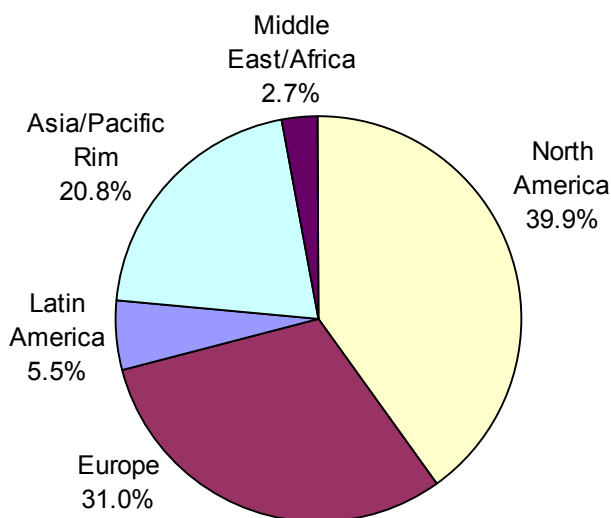
Chapter Four:

Global IT Spending Outlook

In 2002, North America, Europe, and Asia Pacific Rim will comprise 91.7% of the world's IT spending.

Aberdeen's global IT spending forecast for 2002 puts total worldwide IT expenditures at \$1.24 trillion. Growing at a Compound Annual Growth Rate (CAGR) of 4.3% through 2005, Aberdeen predicts worldwide IT spending to reach \$1.43 trillion by that time. Although certain sectors will exceed this growth rate, Aberdeen research indicates that overall long-term growth is limited by GDP growth and corporate top-line revenues. Not surprisingly, North America leads all other regions in spending, followed by Europe and Asia Pacific Rim (Figure 1). These three regions lead all others by a substantial margin and are the most significant established markets.

Figure 1: 2002 World Information Technology Spending

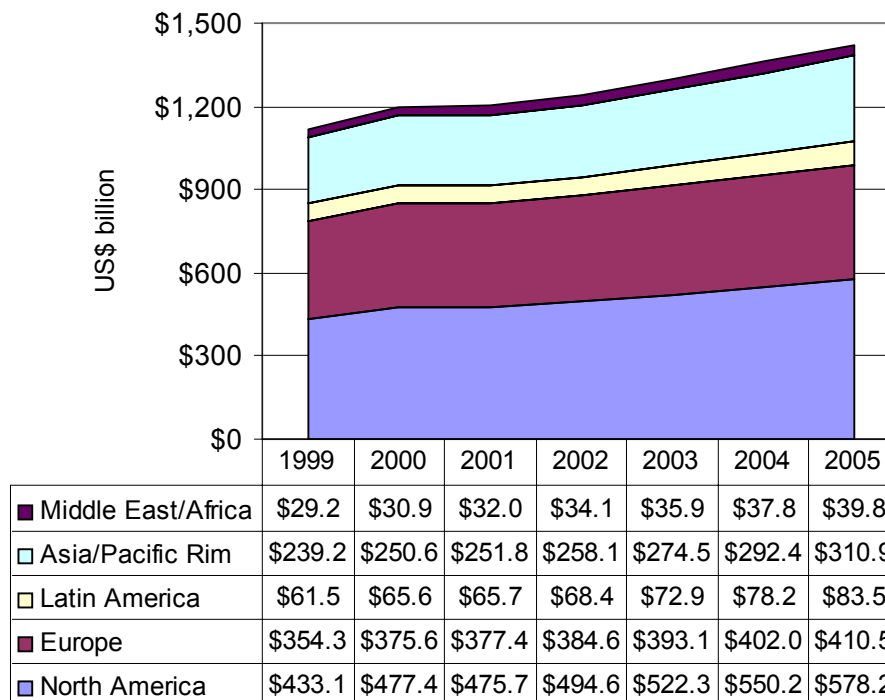


Source: Aberdeen Group, March 2002

IT spending in the three leading regions — North America, Europe, and Asia Pacific Rim — will be led by Asia Pacific Rim at a CAGR of 5.4%, followed by North America at a CAGR of 5.0%, and Europe at a CAGR of 2.1% for the period. Overall spending growth is likely to be slightly higher in the Asia Pacific Rim region as more countries ramp-up IT spending to a level closer to their geographic peers, with a significant contribution from China. In 2002, North America, Europe, and Asia Pacific Rim will comprise 91.7% of the world's IT spending. This share will change minimally. In 2005, these three regions will still make up the lion's share (91.3%) of total global IT spending.

Although Asia will experience slightly higher growth rates in IT spending over the next four years, North America, driven principally by the U.S., will maintain its position as the world's leading region for IT spending. With a CAGR around 5.0% for the period 2001 – 2005, North America will spend \$578 billion on IT by 2005 (Figure 2). Europe's CAGR in the same period will be lower, at 2.1%, and by 2005 Europe will be spending \$410 billion on IT. Asia Pacific Rim will experience the highest growth rate of the three leading regions, with a CAGR of 5.4%. By 2005, it will be spending \$311 billion on IT.

Figure 2: 1999-2005 World IT Spending



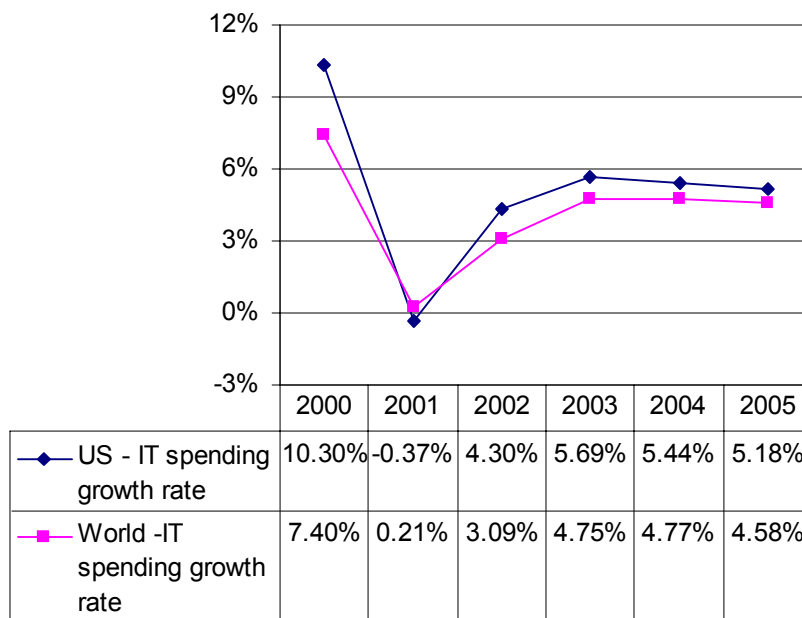
Source: Aberdeen Group, March 2002

Growth Estimates

As seen in Figure 3, Aberdeen forecasts that world IT spending will resume growth in 2002. This will be led by an upswing in U.S. spending. With a 37.1% share of global IT spending, the U.S. is the dominant factor in determining the future of the industry.

Aberdeen does not expect IT spending to resume the double-digit growth experienced in the second half of the 1990's and the year 2000. Aberdeen believes this market-wide hyper-growth was an anomaly that could not be sustained. With IT spending in the U.S. totaling 4.65% of the GDP, technology budgets can no longer run wild and must be viewed in the same vein as other expenditures. As a result, we believe that long-term IT spending increases must come closer to the GDP and corporate revenue growth.

Figure 3: World and U.S. IT Spending Growth Rate



Source: Aberdeen Group, March 2002

Table 1 shows the growth rates for the 10 largest IT markets for the period 2000 to 2005. As shown, the second and third largest markets, Japan and Germany, respectively, will exhibit negative growth in 2002 and will only recover slowly in the 2003 to 2005 period. As a result, total IT spending in 2005 for these two markets will be virtually the same as their levels in 2001 (see Figure 4).

Table 1: IT Spending Growth Rate - Countries
Ranking 1st to 10th

	Country	2000	2001	2002	2003	2004	2005
1	United States	10.30%	-0.37%	4.30%	5.69%	5.44%	5.18%
2	Japan	1.31%	-3.59%	-3.62%	1.28%	1.18%	1.07%
3	Germany	4.63%	-3.44%	-1.15%	1.03%	0.92%	0.81%
4	U.K.	4.63%	2.78%	1.87%	2.72%	2.56%	2.40%
5	France	5.74%	2.32%	0.57%	0.16%	0.08%	0.00%
6	China	16.78%	12.13%	11.59%	14.49%	13.84%	13.18%
7	Italy	4.78%	3.88%	3.03%	1.51%	1.45%	1.38%
8	Canada	9.34%	-0.21%	-0.94%	4.05%	3.85%	3.64%
9	Brazil	8.62%	3.88%	4.39%	6.35%	6.07%	5.78%
10	Mexico	14.38%	1.60%	2.60%	9.21%	8.80%	8.38%

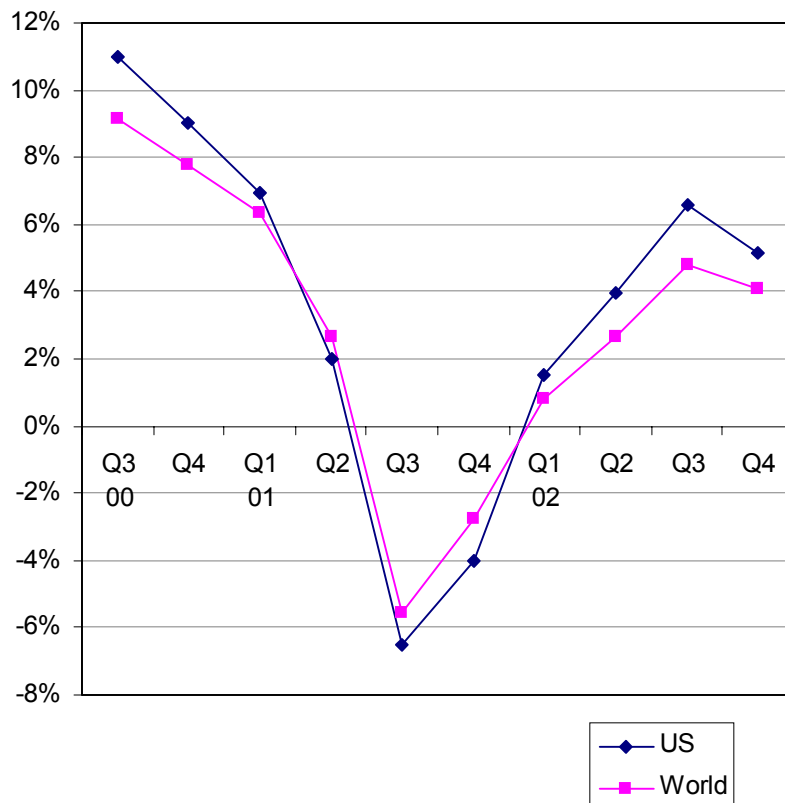
Source: Aberdeen Group, March 2002

Quarterly Growth

Aberdeen expects IT spending to show slight growth in first quarter 2002, although the real recovery will not begin until second quarter 2002.

Figure 4 breaks out IT spending growth by quarters, comparing each period with the same one in the previous year. As can be seen, IT spending growth began to decline in late 2000 and continued through third quarter 2001. In fourth quarter 2001, IT spending growth was still negative, although off the lows experienced in the previous quarter, which saw a decline at an annualized rate of 6.5% for the U.S. and 5.6% worldwide. Aberdeen expects IT spending to show slight growth in first quarter 2002, although the real recovery will not begin until second quarter 2002. It is important to note that the high rate of growth forecasted in third quarter 2002 is in part due to the dreadful performance in third quarter 2001, against which that period is compared.

Figure 4: IT Spending - Quarterly Growth Rates



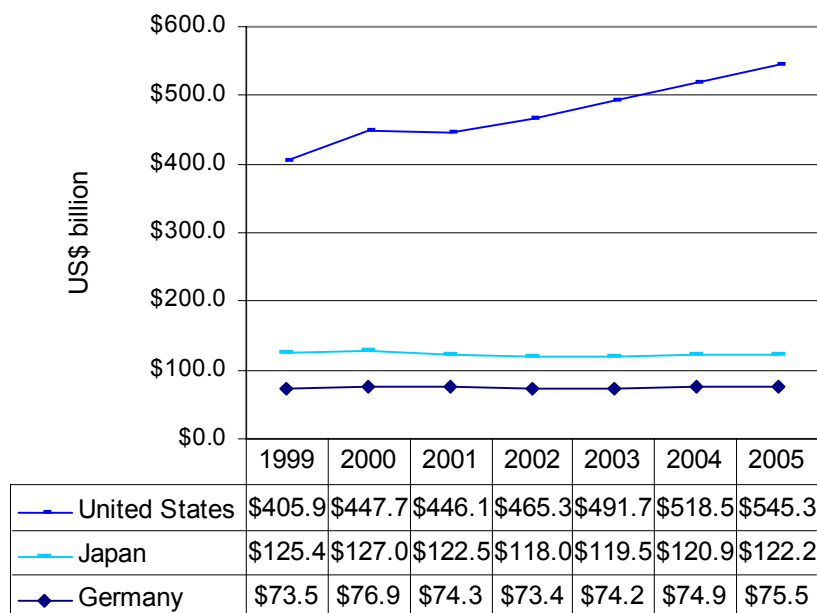
Source: Aberdeen Group, March 2002

Table 2: Quarterly IT Spending Table

	Q1 01	Q2 01	Q3 01	Q4 01	Q1 02	Q2 02	Q3 02	Q4 02
United States	\$110.7	\$123.9	\$112.1	\$111.6	\$103.5	\$118.9	\$113.8	\$116.0
World	\$296.6	\$332.0	\$298.7	\$301.0	\$280.1	\$322.8	\$301.1	\$309.0

Top Country Spending

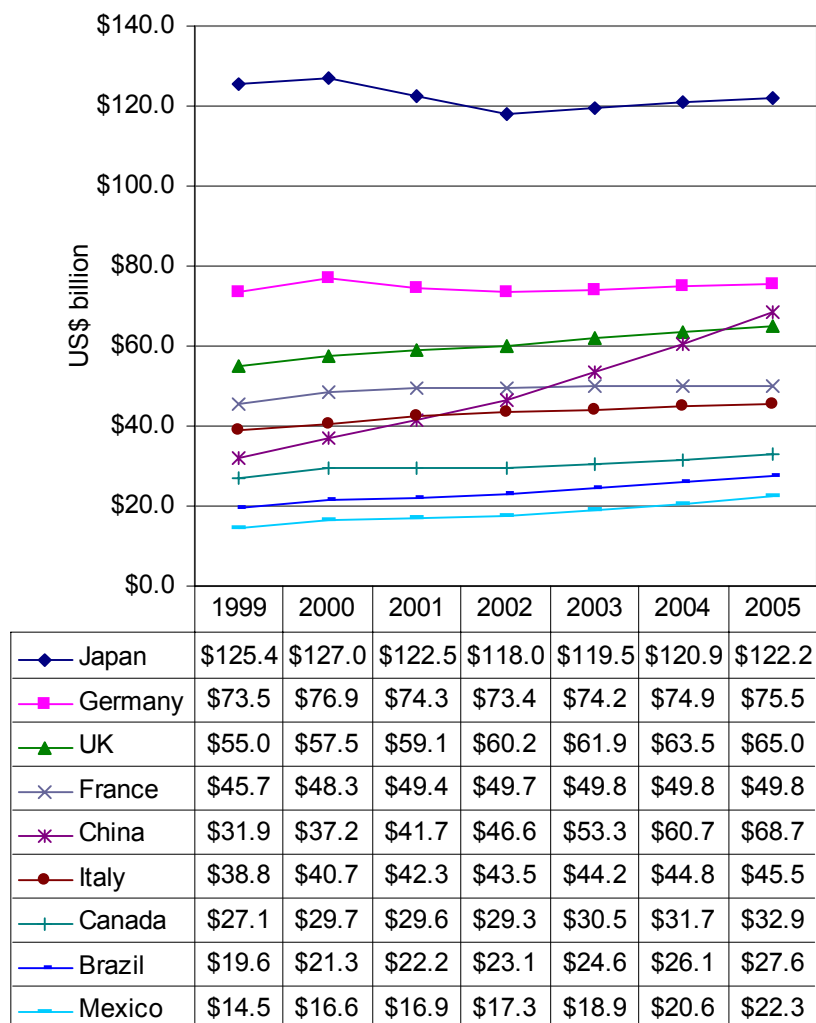
In Figure 5, the total IT spending for the three largest countries is shown. As shown, the U.S. dominates at over three times the expenditures of the nearest country, Japan.

Figure 5: IT Spending in the Three Largest Markets 1999 - 2005

Source: Aberdeen Group, March 2002

In Figure 6, Aberdeen eliminated the U.S. in order to show the other top 10 countries in greater detail. The most significant megatrend is China's transition from the number seven spot in 2000 to the number four position in 2005. This is driven by a CAGR of 13.5% from 2001 to 2005.

Figure 6: IT Spending — Countries Ranking 2nd to 10th

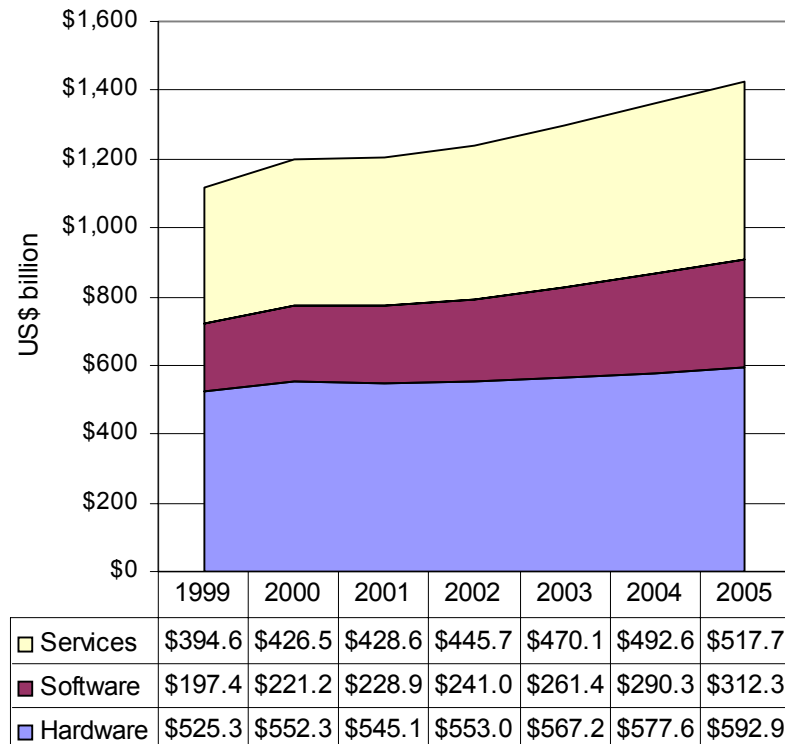


Source: Aberdeen Group, March 2002

Hardware, Software, and Services

For 2001 to 2005, hardware growth will be eclipsed by expansion in software and services spending (Figure 7). Software sales will experience a CAGR of 8.1%, while services will increase at a CAGR of 4.8% and hardware will grow at a CAGR of 2.1%. Or, from a different perspective, suppliers over the same timeframe are competing over \$83.4 billion in additional software spending, \$89.1 billion in new services revenues, and only \$47.8 billion in additional hardware spending.

Figure 7: World Hardware, Software, and Services Spending 1999 - 2005



Source: Aberdeen Group, March 2002

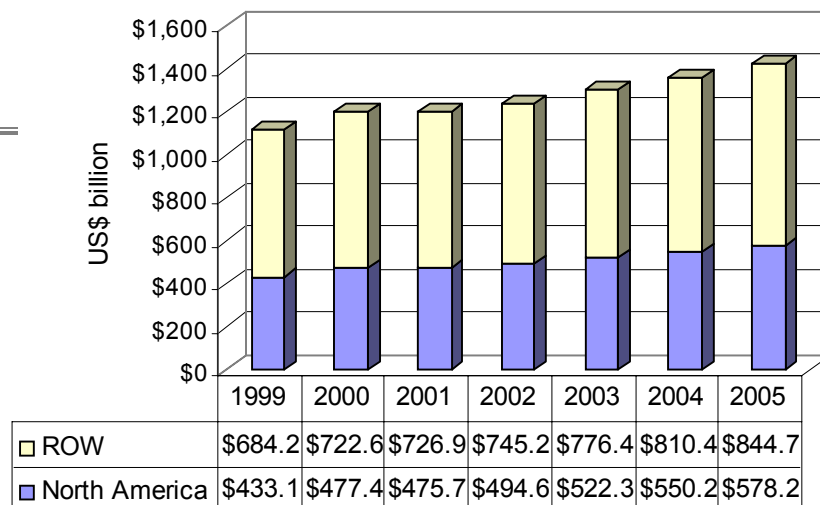
Chapter Five: North America

Overview

In 2001, the North America region represented 39.6% of total global IT spending; that percentage will grow to 40.6% in 2005 (Figure 8).

American IT spending will grow from \$446.1 billion in 2001 to \$545.3 billion by 2005 with a CAGR of 5.15%.

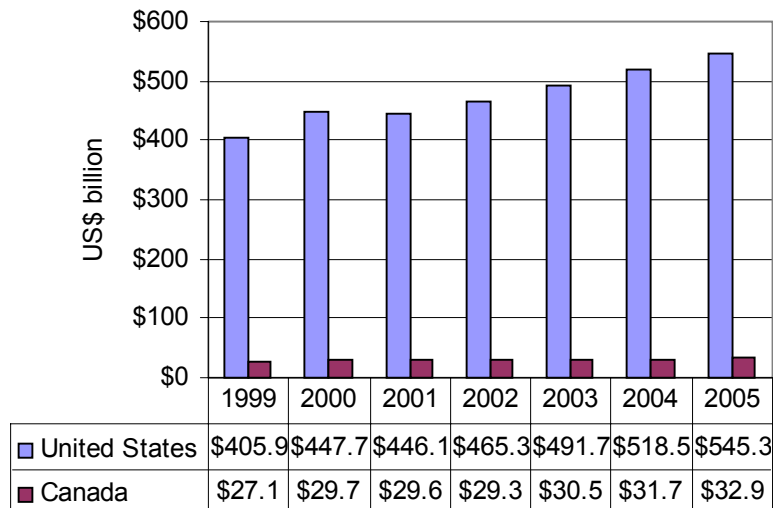
Figure 8: 1999 - 2005 IT Spending – North America



Source: Aberdeen Group, March 2002

The U.S. drives North American IT spending, as Canada's contribution to regional totals will remain roughly 6% through 2005. Easily the world's largest economy, American IT spending will grow from \$446.1 billion in 2001 to \$545.3 billion by 2005 with a CAGR of 5.15% (Figure 9). Canada's IT spending will also grow, but at a lower CAGR of 2.63% for 2001 to 2005.

From a developmental perspective, Canada and the U.S. are two of the world's most advanced countries. The difference in their contributions to global IT spending can be attributed to GDP differentiation between the two countries.

Figure 9: 1999 - 2005 IT Spending by Country – North America

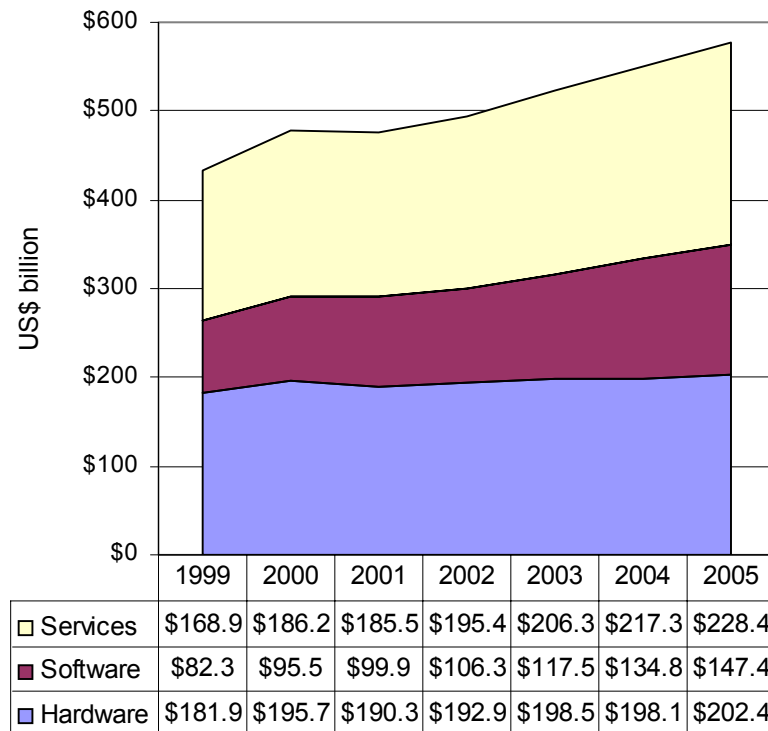
Source: Aberdeen Group, March 2002

Spending Mix

The hardware, software, and services breakdown for North America reveals an interesting trend that will track across well-developed regions and countries. The trend in question is the rise in software and services spending relative to hardware spending (Figure 10). In North America in 2001, hardware sales account for approximately 40% of all IT spending, and services spending makes up a further 39%, leaving software with 21% of all regional IT spending. Hardware's share of total regional IT spending will steadily decrease through 2005 to approximately 35%. During this period, both software and service's percentage of regional IT spending will increase and, by 2005, the largest share of North American IT spending will be 39.5% for services. Software will have increased to 25.5%. Due to the favorable overall trend in spending through 2005, revenue for all sectors will increase, with a CAGR of 1.6%, 10.2%, and 5.3% for hardware, software, and services, respectively.

The slowing growth of hardware spending can be attributed to the fact that enterprises will continue to seek ways to leverage their already existing infrastructure, as well as continued improvements in price/performance. Continually increasing software sales and a need to integrate a variety of applications — both new and old — with a wide mix of hardware platforms will drive services sales. Software sales will increase as organizations seek to take advantage of new business applications, particularly in the mid-market.

Figure 10: 1999 - 2005 Hardware, Software, and Services Spending – North America



Source: Aberdeen Group, March 2002

Chapter Six:

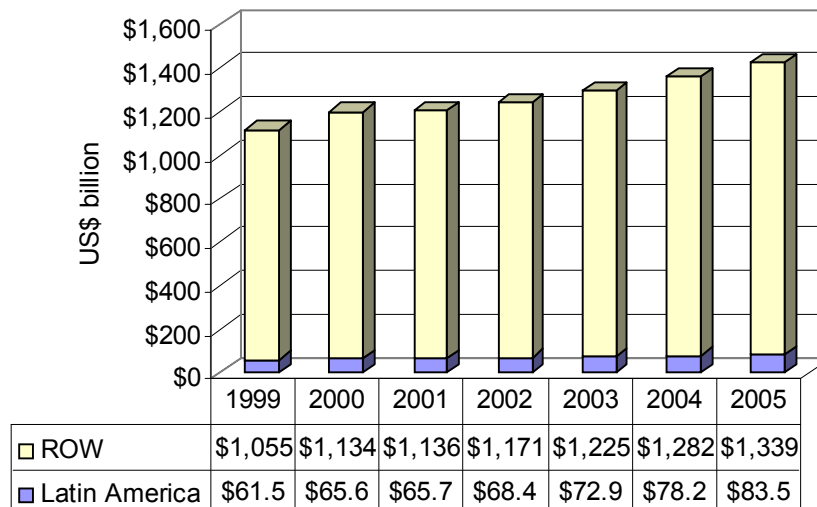
Latin America

Overview

As a region, Latin America spent \$65.7 billion on IT in 2001, and this figure is expected to increase by 6.17% to \$83.5 billion by 2005 (Figure 11). This growth rate is the highest among all the regions covered in this report. Latin America's share of worldwide IT spending will grow from 5.5% in 2001 to 5.87% in 2005. Though many countries in the region are still in the early to middle stages of development, significant regional players are emerging globally and will drive IT spending. In addition, continued success by some of the region's smaller countries in implementing economic stability will further encourage IT spending growth as countries and enterprises strive to modernize in order to capture greater market share outside the region.

Though many countries in Latin America are still in the early to middle stages of development, significant regional players are emerging globally and will drive IT spending.

Figure 11: 1999 - 2005 IT Spending – Latin America



Source: Aberdeen Group, March 2002

Growth in this region will be led by Brazil and Mexico, the two countries characterized by the highest IT spending and by positive CAGRs of 5.64% and 7.21%, respectively, for the period 2001 – 2005 (Figure 12).

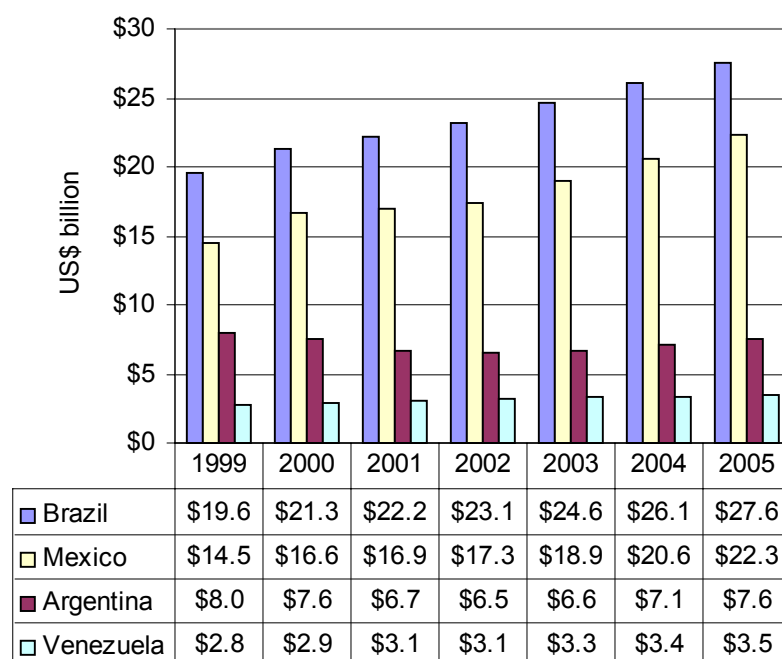
Brazil will retain its position at the top of this region through 2005, increasing its IT spending from \$22.2 billion in 2001 to \$27.6 billion

in 2005, at a CAGR of 5.64% for the period. The country's successful stabilization of the economy over the last decade and its highly developed mining, service, manufacturing, and agriculture sectors will drive its IT spending through 2005.

Mexico's high IT spending growth will be driven by high oil prices, exports, higher wages, and growing consumption. Add to that picture the doubling of trade with the U.S. and Canada since NAFTA (North American Free Trade Agreement) was signed. IT spending in Mexico will grow from \$16.9 billion in 2001 to \$22.3 billion in 2005.

Argentina will suffer short-term from its economic crisis, but will slowly return to growth in 2003, which will accelerate in 2004 and 2005. Venezuela will experience moderate growth, increasing from \$3.1 billion in 2001 to \$3.5 billion in 2005.

Figure 12: 1999 - 2005 IT Spending by Country – Latin America



Source: Aberdeen Group, March 2002

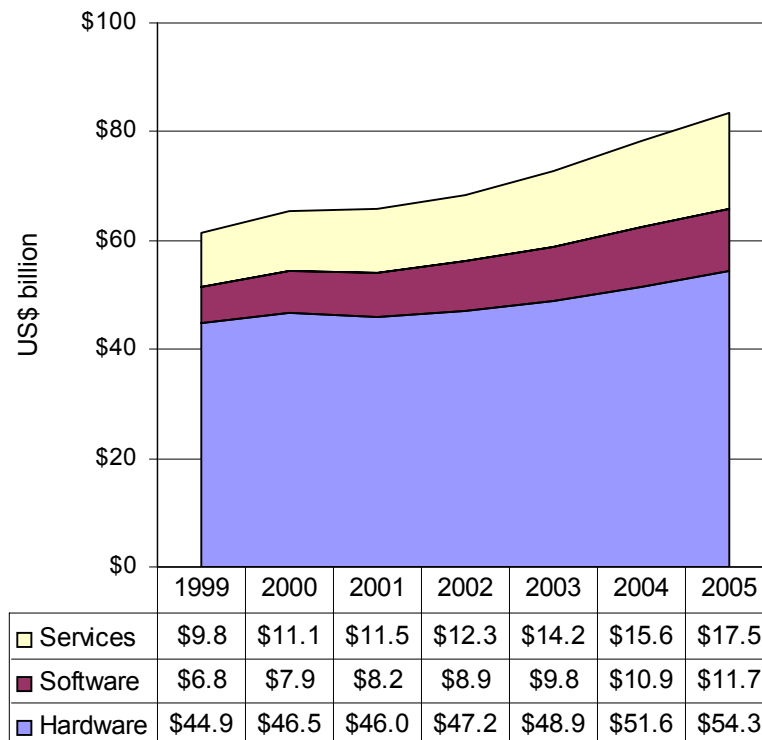
Spending Mix

As a region, Latin America's IT spending is dominated by hardware sales, which consist of nearly 70% of 2001 spending at \$46 billion (Figure 13). Although hardware's share of total regional spending will decrease over four years to 65%, it will still dominate. During the 2001 – 2005 period, software and services spending will grow at

CAGRs of 9.22% and 11.12%, respectively. Hardware spending will progress at a slower rate of 6.17%. In 2005, software spending as a share of the regional total will have grown from 12.5% to 14%, and the services share of spending will have increased from 17.5% to 21%. Combined, these two segments will account for just under \$30 billion in IT spending by 2005.

Although hardware sales in Latin America will remain very healthy at over \$54 billion in 2005, software and services spending will see marked increases due primarily to the growing need for systems integration and implementation services.

Figure 13: 1999 - 2005 Hardware, Software, and Services Spending – Latin America



Source: Aberdeen Group, March 2002

Chapter Seven:

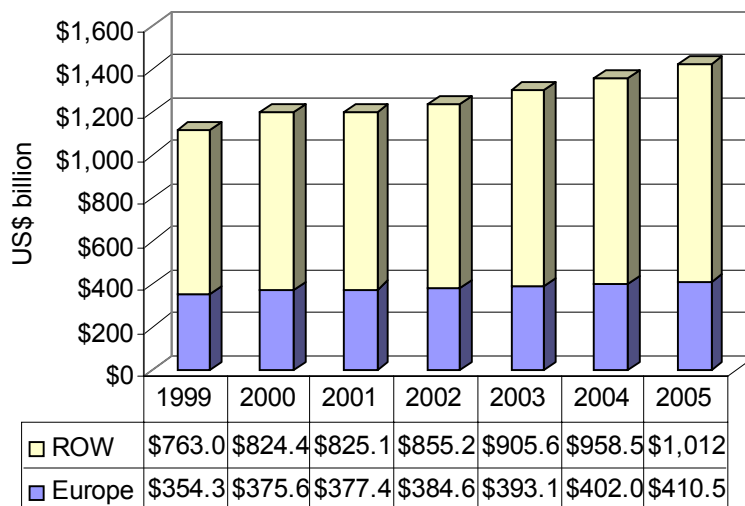
Europe

Overview

The world's second leading region in terms of both GDP and IT spending with a large number of constituent countries, Europe contains a greater variety of spending mix and market opportunity.

European IT spending will grow at a CAGR of 2.12% between 2001 and 2005, from \$377.4 billion to \$410.5 billion (Figure 14). The world's second leading region in terms of both GDP and IT spending with a large number of constituent countries, Europe contains a greater variety of spending mix and market opportunity. With few exceptions, the countries that make up Europe are highly industrialized and stable. In addition, they already have high levels of IT infrastructure in place, regardless of their economy's size.

Figure 14: 1999 - 2005 IT Spending – Europe



Source: Aberdeen Group, March 2002

European countries, ranking first to sixth for IT expenditures, spent from \$17.4 billion to \$74.29 billion in 2001, a range that will grow to between \$19.6 billion and \$75.48 billion in 2005 (Figure 15). European total IT spending will grow at a CAGR of 2.12% from 2001 to 2005. All countries' IT spending in this group will grow at rates ranging between 0.2% (France) and 2.97% (the Netherlands).

European IT spending patterns are based on its level of economic sophistication and its highly developed private sector as well as high levels of consumer spending.

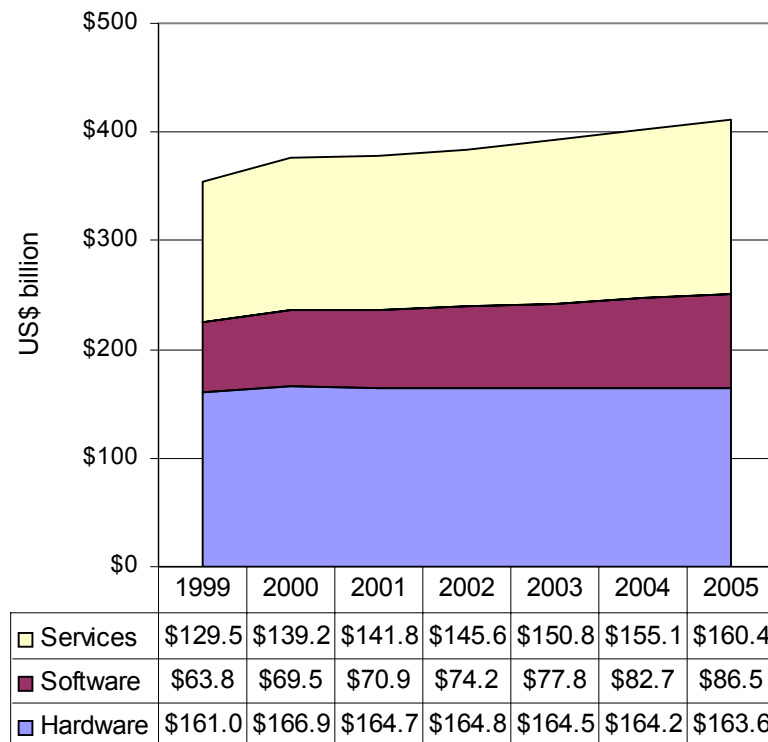
Figure 15: 1999 - 2005 IT Spending by Country — Europe

Source: Aberdeen Group, March 2002

Spending Mix

Europe's mix of hardware, software, and services spending mirrors that of North America, due in large part to the developed nature of the region as a whole. In 2001, hardware represented a 43.6% share of total IT spending in the region, followed by services with 37.6%, and software rounding out the three with the remaining 18.8% (Figure 16). Over the next four years, services and software will each gain market share at the expense of hardware. This trend is likely to accelerate beyond 2005, closely following the ratio change observed in North America during 2001 – 2005 whereby services will eventually become the largest sector.

Figure 16: 1999 - 2005 Hardware, Software, and Services Spending — Europe



Source: Aberdeen Group, March 2002

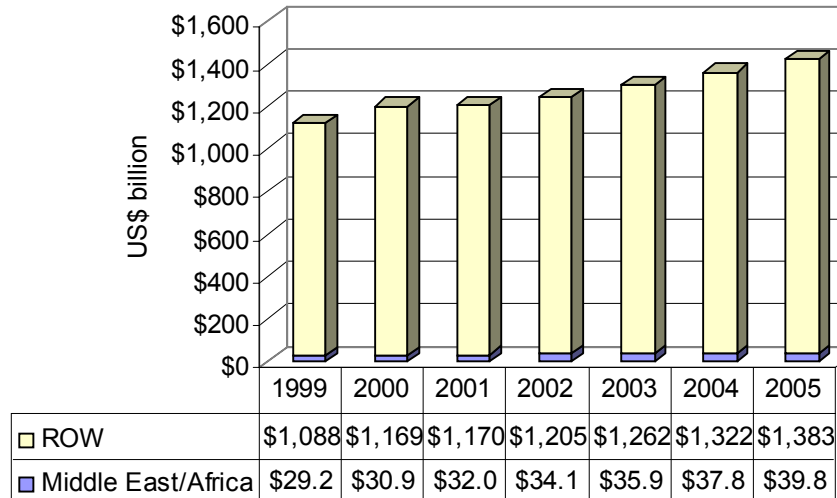
Chapter Eight:

Middle East and Africa

Overview

The Middle East and Africa region's IT spending will increase at a CAGR of 5.61% from \$32 billion to \$39.8 billion between 2001 and 2005 (Figure 17). From an IT adoption perspective, Middle East and Africa lags behind other areas of the world.

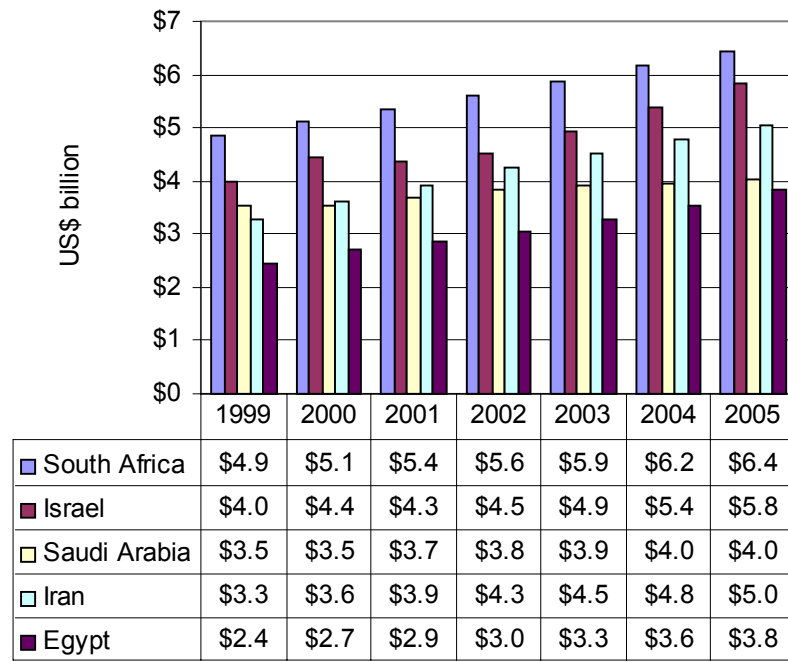
Figure 17: 1999 - 2005 IT Spending — Middle East and Africa



Source: Aberdeen Group, March 2002

Egypt's IT spending will experience marked growth in the period 2001 – 2005, increasing from \$2.9 billion to \$3.8 billion (Figure 18). Egypt's IT spending growth through 2005 is the result of successful implementation by the government of IMF (International Monetary Fund) plans to improve the economy and stabilize monetary policy. Egypt's external debt, at one time extremely large, has been drastically reduced in return for acting as a coalition partner in the Gulf War in 1990. Additionally, although tourism and Suez Canal revenues have dropped since the late 1990's, the resurgence of oil prices has boosted its GDP and improved overall economic health.

Israel is the most well-developed country in the group, especially with regard to IT indicators such as IP addresses, PCs, and telephone lines. Israel's lower IT spending growth rate is more in line with other countries that are similarly well-developed.

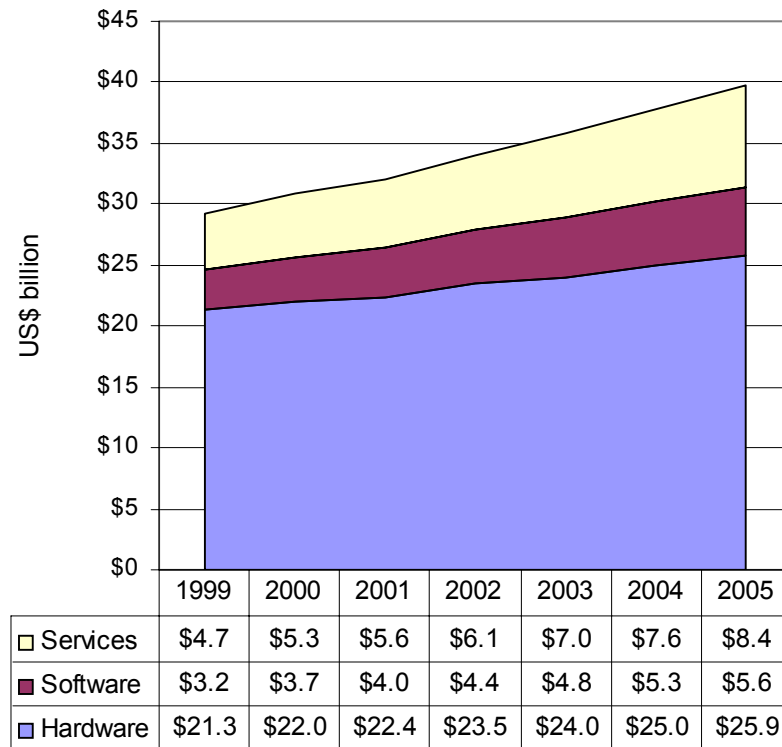
Figure 18: 1999 - 2005 IT Spending by Country — Middle East and Africa

Source: Aberdeen Group, March 2002

Spending Mix

The IT spending mix for the Middle East and Africa region is representative of the developing nature of most of the region's economies. Hardware spending will increase from \$22.4 billion in 2001 to \$25.9 billion in 2005, growing at a CAGR of 3.67% (Figure 19). During that period, hardware (as a share of total IT spending) will decrease from 70% in 2001 to 65% in 2005. Hardware's dominance of the Middle East and Africa IT spending market remains relatively unchallenged through 2005. Software and services will both grow at CAGRs of 8.6% and 10.5%, respectively, between 2001 and 2005. For the region, software will grow from \$4.0 billion in 2001 to \$5.6 billion in 2005. Services will increase from \$5.6 billion in 2001 to \$8.4 billion in 2005.

Figure 19: 1999 - 2005 Hardware, Software, and Services Spending — Middle East and Africa



Source: Aberdeen Group, March 2002

Chapter Nine:

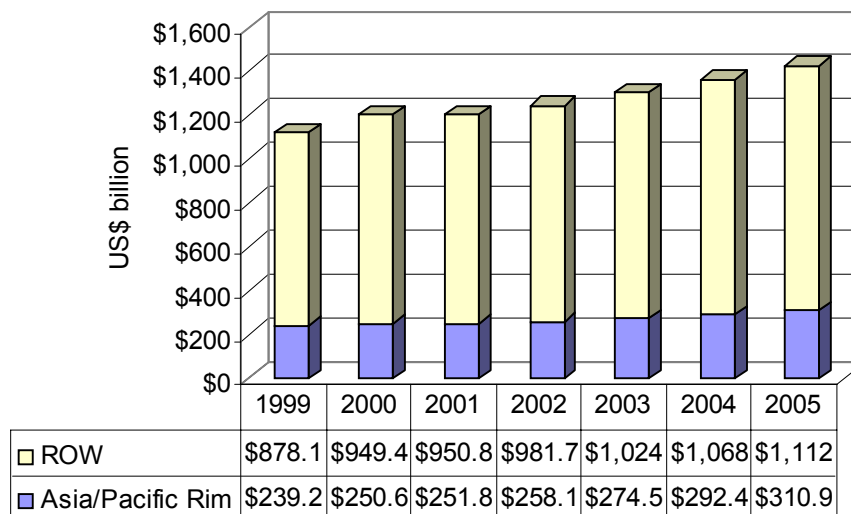
Asia and Pacific Rim

Overview

The Asia Pacific Rim (APR) region's IT spending will grow from \$251.8 billion in 2001 to \$310.9 billion in 2005.

The Asia Pacific Rim (APR) region's IT spending will grow from \$251.8 billion in 2001 to \$310.9 billion in 2005 (Figure 20), with a CAGR of 5.4% for the period. APR's share of worldwide IT spending will increase from 20.94% in 2001 to 21.85% by 2005. As a region, APR has a very diverse constituency and correspondingly diverse growth prospects. Not surprisingly, however, the region's most developed countries do not differ from the IT spending patterns identified for North America and Europe.

Figure 20: 1999 - 2005 IT Spending — Asia and Pacific Rim



Source: Aberdeen Group, March 2002

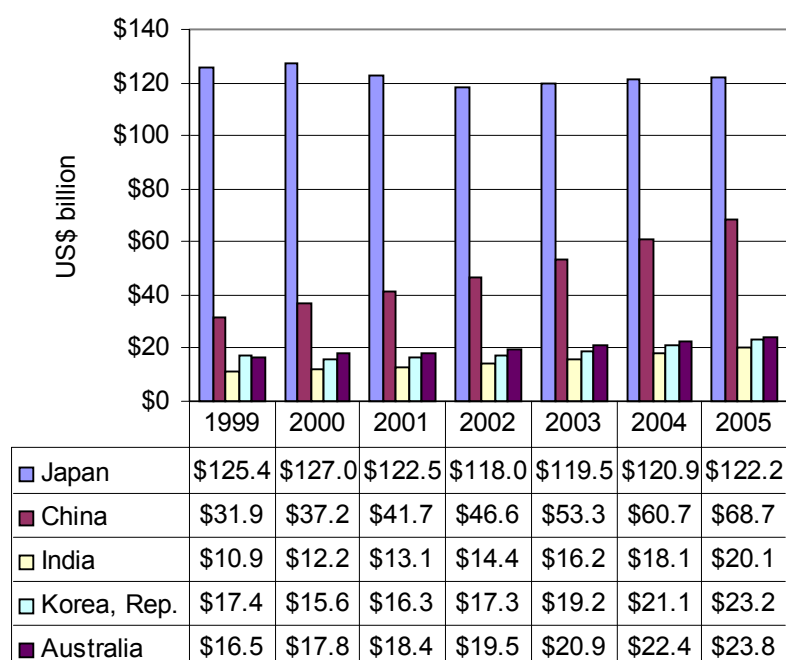
APR's growth will be led by China and India, whose IT spending will grow at a CAGR of 13.27% and 11.25%, respectively, between 2001 and 2005. Korea and Australia will follow with CAGRs of 9.23% and 6.76%, respectively (Figure 21).

China and India are poised for explosive IT spending growth over the next four years, a fact that should come as no surprise. India has long been an exporter of highly educated IT employees and, due to the low cost of labor, has become the seat of many new ventures as more of those highly educated citizens choose to stay home or return to India. Similarly, China's steady trend toward a more market-

oriented economy has met with success and, although government control remains tight, newly privatized government corporations and the rapidly increasing private sector based on the country's coast will fuel increased IT spending. In 2001, the two countries together accounted for 21.8% of this region's IT spending — this share will grow to 28.6% by 2005. Paradoxically, both India and China are home to vast rural and impoverished populations whose exclusion from this boom may create tension worth monitoring. In addition to social issues, both India and China are major players in several long-standing, international disputes that could affect spending.

Japan is characterized by the highest IT spending in the region (and the second largest IT spending in the world) but also by the lowest IT spending CAGR of any country in the APR — at 0.05% between 2001 and 2005 due to its continuing economic distress and lack of capital investment. Australia and Korea will remain important IT markets over the next four years. Japan, Korea, and Australia represent more well-established and developed markets in the APR region.

Figure 21: 1999 - 2005 IT Spending by Country — Asia and Pacific Rim

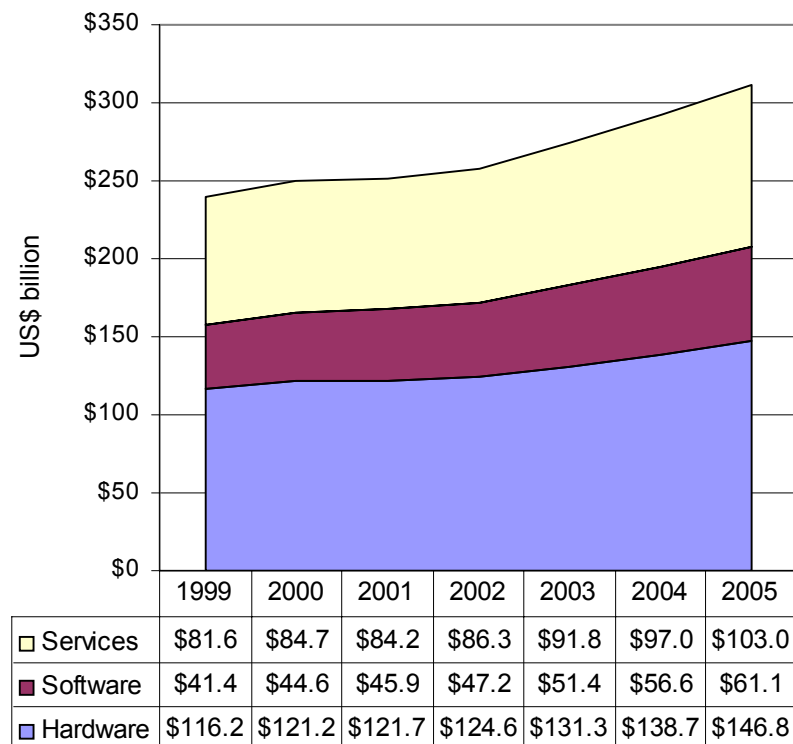


Source: Aberdeen Group, March 2002

Spending Mix

The hardware, software, and services mix of IT spending for APR will reflect the region's overall composition of developed and developing economies. Although APR ranks a close third in overall regional IT spending to North America and Europe, the region contains far more developing economies than either of the other two. Consequently, as observed throughout this report, hardware will continue to dominate IT spending, increasing from \$121.7 billion in 2001 to \$146.8 billion in 2005 (Figure 22). Yet, following other observed patterns, software and services spending in particular will increase their share of overall regional IT spending. Software will grow from \$45.9 billion in 2001 with an 18.2% share of regional IT spending to \$61.1 billion in 2005 with a 19.6% share of spending. Services will grow to a \$103 billion market in 2005.

Figure 22: 1999 - 2005 Hardware, Software, and Services Spending — Asia and Pacific Rim



Source: Aberdeen Group, March 2002

Author Profiles

Hugh Bishop

As Senior Vice President of Aberdeen's Emerging Technology Intelligence (ETI) Group, Hugh Bishop directs Aberdeen's ongoing qualitative and quantitative coverage for new technology markets. As part of these activities, Bishop directs Aberdeen's Technology Forecasting Consortium (TFC), an end-user advisory panel that assists Aberdeen in early identification of trends in the Information Technology market and prediction of user buying intentions. Additionally, Bishop heads up Aberdeen's quantitative research, providing market forecasts and supplier market share estimates on a worldwide basis. Bishop's recent work includes *Technology Forecasting Consortium 2002: User Buying Intentions* (February 2002), an Aberdeen Research Report.

Prior to his current activities, Bishop was Senior Vice President of Aberdeen's Enterprise Business Applications practice. Additionally, Bishop was the founder and lead analyst of Aberdeen's Customer Relationship Management (CRM) practice.

Prior to joining Aberdeen, Bishop worked at Ryan Associates, a marketing consulting firm. He has also served as Director of Computer Operations at a leading, Boston-based engineering firm.

Esmé Fantozzi

Esmé Fantozzi is a senior research associate with Aberdeen's Emerging Technology Intelligence (ETI) group. She identifies and tracks critical and cutting-edge information specific to hyper-growth and emerging Information Technology (IT) and high-tech markets.

Fantozzi coauthored *Optical Circuits: Market Realities & Potential— The Evolution of the Planar Lightwave Circuit (PLC) Market* (October 2001), an Aberdeen Research Report that provides in-depth analysis of market and technology trends related to integrated optical components. Fantozzi also coauthored *Net Market Technology: Internet Hubs Drive Business Process Reengineering* (March 2001), an Aberdeen Research Report that helps IT executives, technology suppliers, and investors navigate, comprehend, and capitalize on the benefits of the Net market space. Fantozzi also collaborated on *Strategic e-Sourcing: A Framework for Negotiating Competitive Advantage* (April 2001), an Aberdeen Research Report that examines the most direct and effective ways that an organization can utilize the Internet to control costs, improve profits, and enhance overall responsiveness to changing market dynamics.

Fantozzi brings several years of research, consulting, and international product management experience to Aberdeen Group. Prior to joining Aberdeen, she completed a feasibility study on innovative energy sources for a Fortune 100 company. As a physics researcher at Pennsylvania State University, Fantozzi designed and implemented an advanced data acquisition system. In Europe, Fantozzi held a Product Manager position at a General Electric company.

Fantozzi holds an M.S. and a B.S. in Physics with a specialization in Electronics from the University of Padua in Italy.

Appendix A

Lexicon of Acronyms and Abbreviations

APR	Asia Pacific Rim
CAGR	Compound Annual Growth Rate
GDP	Gross Domestic Product
IMF	International Monetary Fund
IT	Information Technology
NAFTA	North American Free Trade Agreement
ROW	Rest of World
WTSM	World Technology Spending Model

Appendix B

Related Aberdeen Research

Aberdeen Group has produced several publications that provide complementary market research and strategic market information relating directly to this report:

- Technology Forecasting Consortium: 2002 User Buying Intentions” (February 2002);

Information on these and any other Aberdeen publications can be found at www.aberdeen.com or by e-mail at info@aberdeen.com.